

Copulas & Tail Dependence

Correlation is a photograph taken in calm weather. The corner of the joint distribution is where the money is lost.

A correlation matrix is a photograph of dependence taken in calm weather. The question risk managers actually get paid to answer — **if the S&P has one of its worst weeks, what happens to London and Tokyo?** — lives in the corner of the joint distribution, where a single number cannot see. Copulas separate what each market does on its own from how they move together, and once you make that split on 25 years of real index data, the verdict is blunt: the Gaussian copula says joint crashes are vanishingly rare; the data says they cluster; and a Student-t copula with 3.5 degrees of freedom repairs most of the damage.

PROJECT CARD

CATEGORY

Risk Management

LIBRARIES

SciPy · NumPy · Pandas

ASSETS

^GSPC, ^FTSE, ^N225

TIMEFRAME

Jan 2000 – Dec 2024

USE

Stress-testing a multi-asset book against the joint tail, not the average

- 01

Weekly log returns for ^GSPC, ^FTSE, ^N225, 2000–2024 (holiday-aligned)
- 02

Pearson vs Spearman vs Kendall — three answers to ‘how correlated?’
- 03

Rank-transform to pseudo-observations on the unit square
- 04

Fit Gaussian and Student-t copulas (df by MLE over a 2–30 grid)

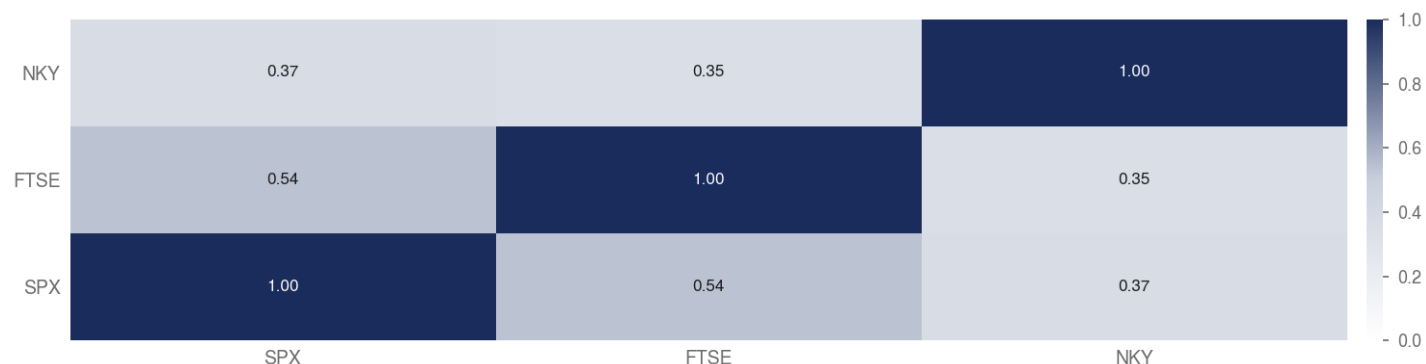
1. Dependence is not correlation

The sample is 1,302 aligned weekly returns for the S&P 500, FTSE 100 and Nikkei 225, 2000-01-03 to 2024-12-31 — dot-com bust, 2008, the euro crisis, COVID and the 2022 rates shock all included. Before any statistics, two traps come free with global indices, and both are worth learning to spot. First, each exchange keeps its own holiday calendar, so returns are computed jointly and any week missing a market is dropped. Second — subtler — the markets do not even trade at the same time. Tokyo closes roughly ten hours before London and some fifteen before New York, so a same-calendar-day Nikkei close **leads** the US close, and daily cross-market correlations are structurally understated: the two prices are answering questions asked at different moments. Resampling to weekly (Friday-to-Friday) returns absorbs most of that offset — watch the SPX–Nikkei Pearson correlation jump from 0.15 on daily data to 0.58 on weekly.

Even then, “the correlation” is three different numbers. Pearson measures linear co-movement and is hostage to outliers; Spearman and Kendall are rank-based, so they see only the ordering — exactly the part a copula models. Kendall’s τ has the cleanest interpretation: the probability that two randomly chosen weeks agree in direction, minus the probability they disagree.

Kendall’s τ , weekly returns 2000–2024

Rank correlation: SPX–FTSE 0.54, SPX–NKY 0.37, FTSE–NKY 0.35



Source: Author calculations, from the article’s computed results.

WHY PRACTITIONERS CARE

Every multi-asset stress scenario is implicitly a statement about joint tails. If your scenario engine assumes a dependence structure with no tail clustering, diversification will look better on paper than it behaves in a crisis — precisely when you are counting on it.

2. Sklar’s theorem & pseudo-observations

Here is the idea that makes the whole subject click. Sklar’s theorem (1959) says any joint distribution factors cleanly into two parts: the **marginals** — what each market does on its own — and a **copula**, a joint distribution on the unit square that carries all of the togetherness and none of the marginal shape. Separate the dancers from the dance. To see the copula empirically, replace each return by its normalised rank, $u = \frac{\text{rank}(x)}{n+1}$ — the probability integral transform done with the empirical CDF. Ranks are uniform by construction, so the marginals are washed out entirely — any structure that survives in the picture below is pure dependence.

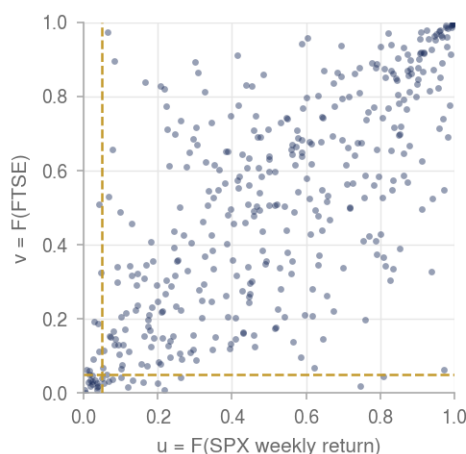
```
from scipy import stats

def pseudo_obs(x):
    # rank / probability integral transform
    return stats.rankdata(x) / (len(x) + 1)
```

```
U = pd.DataFrame({c: pseudo_obs(ret[c].values) for c in ret.columns})
```

Pseudo-observations, SPX vs FTSE — 400 of 1,302 weeks

One point is one week; the dashed lines mark the 5% quantile of each margin



Source: Author calculations, from the article's computed results.

If the two markets were independent this square would be filled uniformly. Instead the mass drains toward the diagonal — and, critically, **piles up in the corners**. The lower-left corner is the object of study for the rest of this article: weeks in which both markets were simultaneously in their worst tail.

3. The Gaussian copula — elegant, and wrong in the corner

The Gaussian copula is what you implicitly assume whenever you summarise joint behaviour with a correlation matrix alone. Fitting it is one line: push the pseudo-observations through the standard normal quantile function (“normal scores”) and take their correlation — for SPX–FTSE that gives $\rho = 0.74$. It is analytically convenient, it scales to any dimension, and it has one fatal property: **zero tail dependence**. Ask it “given one market is having a 1-in- q disaster, how often is the other one too?” and as q deepens its answer marches to zero — for any $\rho < 1$, no matter how

high. In the limit, joint crashes are not just rare. They are assumed away.

You can watch the assumption fail at finite depth. At the 10% level the fitted Gaussian copula implies a joint-crash ratio of 50.8% for SPX–FTSE; at 5% it has slipped to 43.5%; at 1% it is down to 31.2%, on its way to zero. The empirical series goes the other way: 55.3% at 10%, 55.3% at 5%, 61.4% at 1%. The deeper you look into the tail, the more the data diverges from the model — in the direction that hurts.

THE CAUTIONARY TALE

David Li’s 2000 paper “On Default Correlation: A Copula Function Approach” put the Gaussian copula at the centre of CDO pricing, and for a few years a single correlation number priced trillions in structured credit. The formula did exactly what it says here: it made simultaneous defaults look almost impossible at depth. 2008 was, among other things, the empirical tail-dependence estimate arriving.

4. The t copula & tail dependence

The Student-t copula adds exactly one parameter — the degrees of freedom ν — and that single knob buys tail dependence. We set ρ by Kendall's τ inversion, $\rho = \sin\left(\frac{\pi\tau}{2}\right)$, which is exact for elliptical copulas, and profile the exact copula log-likelihood over a ν grid from 2 to 30 on the SPX–FTSE pair:

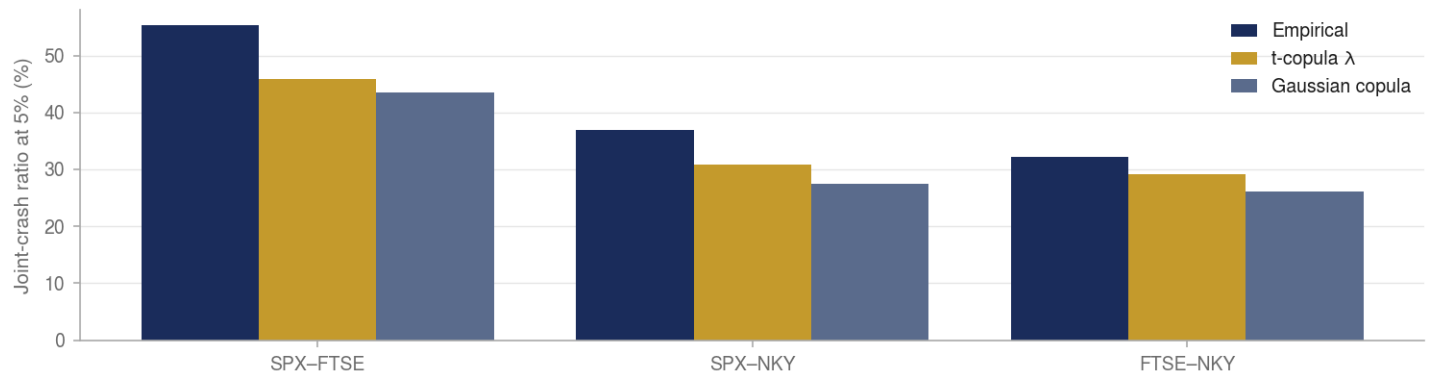
```
rho = np.sin(np.pi * tau / 2)          # tau inversion: 0.746
grid = np.arange(2.0, 30.5, 0.5)
ll = np.array([t_copula_loglik(u, v, rho, df) for df in grid])
df_hat = grid[ll.argmax()]             # -> 3.5

# lower-tail dependence, closed form (Demarta & McNeil 2005)
lam = 2 * stats.t.cdf(-np.sqrt((df_hat+1)*(1-rho)/(1+rho)), df=df_hat+1)
```

The likelihood picks $\nu = 3.5$ — heavy joint tails — and prefers the t copula decisively: log-likelihood 577.9 against the Gaussian's 526.8 on the same pseudo-observations, one extra parameter. Unlike the Gaussian, the t copula's tail dependence does not vanish: it converges to the closed-form λ above, which at $\nu = 3.5$ gives 45.9% for SPX–FTSE, 30.9% for SPX–NKY and 29.2% for FTSE–NKY.

Joint-crash ratio $P(\text{both} < 5\% \text{ quantile}) / 5\%$ — data vs models

The Gaussian bar undershoots the observed ratio for every pair; the t-copula λ is a floor that does not decay



Source: Author calculations, from the article's computed results.

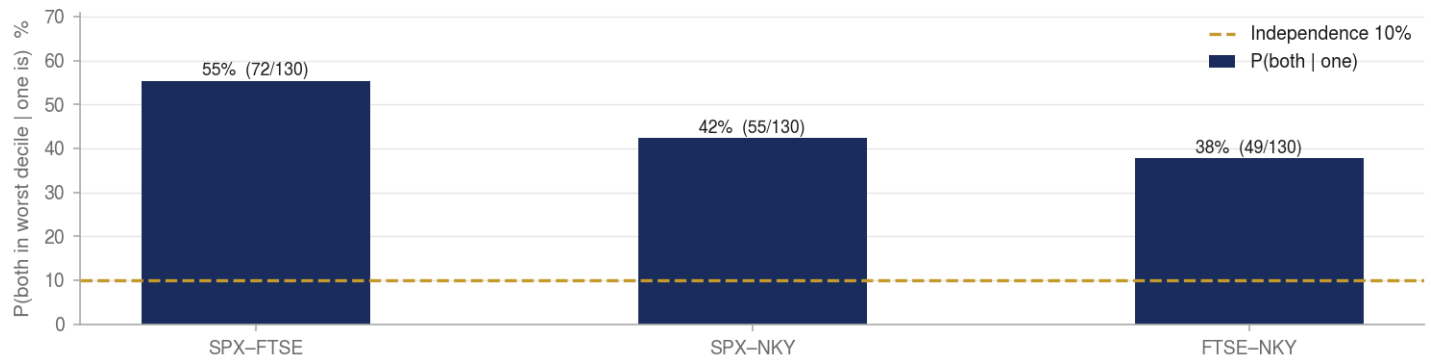
Read the empirical-vs-Gaussian gap per pair: the Gaussian copula, fitted to the **same data** with the **same correlation**, undershoots the observed 5% joint-crash ratio for every pair — and remember its bar keeps falling as q shrinks while the empirical one rises. The middle bar is the t copula's limiting λ : a floor that does not decay, sitting close to what the data shows.

5. Conditional crash probabilities

The same mathematics, phrased the way a risk committee asks it: given that one market has a worst-decile week, what is the probability the other one does too? Under independence the answer would be 10%. The data answers between 37.7% and 55.4%:

P(both in worst decile | one is) per pair — dashed line = the 10% an independence assumption would give

Every pair sits far above the independence line



Source: Author calculations, from the article's computed results.

Pair	P(both worst-decile one is)	Gaussian @10%	t copula λ ($\nu = 3.5$)
SPX-FTSE	55.4% (72/130)	50.8%	45.9%
SPX-NKY	42.3% (55/130)	35.6%	30.9%
FTSE-NKY	37.7% (49/130)	34.1%	29.2%

A worst-decile S&P week drags the FTSE into its own worst decile more than half the time — 72 of 130 of the 130 conditioning weeks — and even the geographically and temporally distant Nikkei follows 42.3% of the time. Note the ordering in every row: empirical $\geq$ t copula $\lambda >$ Gaussian-at-depth. The single-parameter fix gets you most, not all, of the way — the residual gap is the asymmetry between crash and boom corners that symmetric elliptical copulas cannot express.

6. The practitioner take

For multi-asset stress testing, the workflow this tutorial rehearses is the one that survives contact with a crisis:

- **Model marginals separately** — fat tails, volatility clustering, whatever each series needs — then choose the dependence structure as a deliberate act, not as a side effect of writing down a correlation matrix.
- **Check the candidate copula against the data** — compare it to the empirical joint-crash ratios at several depths before trusting it.
- **When in doubt between Gaussian and t, pay the one parameter** — the cheapest tail insurance in the toolbox: here it turned “joint crashes become impossible” into $\lambda \approx 46\%$ for the closest pair.
- **Know where this road ends** — the t copula is elliptical, so its lower-tail dependence comes bundled with an identical upper tail: joint booms priced as generously as joint crashes. When that asymmetry matters, the next tools up are the asymmetric Archimedean families (Clayton glues lower tails only) and vine constructions, which assemble a high-dimensional copula from freely chosen pairs.

PRACTITIONER TAKE

Correlation tells you how markets dance; the copula tells you how they die. Simulate scenarios from a fitted t copula ($\nu = 3.5$ here) rather than a Gaussian one and your stress engine will generate the co-crashing weeks that actually populate the historical record — 130 worst-decile S&P weeks in this sample, the majority of them shared with London.

REFERENCES

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2. Embrechts, P., McNeil, A. & Straumann, D. (2002). Correlation and dependence in risk management: properties and pitfalls. In *Risk Management: Value at Risk and Beyond*, Cambridge University Press.
3. Demarta, S. & McNeil, A. J. (2005). The t copula and related copulas. *International Statistical Review*, 73(1), 111–129.
4. Aas, K., Czado, C., Frigessi, A. & Bakken, H. (2009). Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2), 182–198.
5. Li, D. X. (2000). On default correlation: a copula function approach. *Journal of Fixed Income*, 9(4), 43–54.
6. Companion notebook: copulas - tail - dependence .ipynb — reproduces every figure from raw data (seed 11).